

Shubhadeep Chakraborty

+91-7637895488 | shubhadeep.chakraborty@icts.res.in | Bengaluru, India

github.com/shubhadeep763 | Google Scholar

Physics PhD Candidate | Non-Equilibrium Statistical Mechanics | Hydrodynamics & Transport | Active Matter | Quantum Many-Body Systems

RESEARCH PROFILE

Integrated PhD candidate in Physics at ICTS-TIFR working in non-equilibrium statistical mechanics, with a focus on hydrodynamics and transport in classical and quantum many-body systems, active matter, and stochastic processes. I combine analytical methods — kinetic and hydrodynamic descriptions, macroscopic and fluctuating hydrodynamics, exact diagonalisation, and generalized hydrodynamics — with Python-based numerical and molecular dynamics simulations to connect microscopic dynamics to emergent macroscopic laws.

Research interests: kinetic theory and hydrodynamics of many-body systems; transport in low-dimensional and integrable systems; generalized and fluctuating hydrodynamics; active matter and non-equilibrium steady states; quantum many-body dynamics and quantum walks.

EDUCATION

International Centre for Theoretical Sciences (ICTS-TIFR)

Integrated PhD in Physics

Bengaluru, India

Aug 2023 – Present

St. Xavier's College (Autonomous)

B.Sc. in Physics | SGPA: 8.33 / 10

Kolkata, India

2020 – 2023

INSPIRE Scholar (DST, Government of India)

PUBLICATIONS & PREPRINTS

S. Chakraborty et al., "Hydrodynamics of the Fermi–Pasta–Ulam–Tsingou (FPUT) Chain"

2026

- Preprint, arXiv:2606.04674.
- Derived Euler and Navier–Stokes–Fourier hydrodynamic equations for nonlinear lattice systems from microscopic principles.
- Solved Riemann initial-value problems analytically and verified the predictions through molecular dynamics simulations.

RESEARCH EXPERIENCE

Jamming in a System of Persistent Active Particles in One Dimension

2025 – Present

- Derived exact steady-state cluster statistics by mapping force-balanced configurations to constrained lattice paths using reflection-principle methods.
- Proved scaling behavior for mean jamming time by reducing the many-body dynamics to a first-passage drift–diffusion framework.
- Validated predictions using event-driven molecular dynamics simulations implemented in Python.

Fluctuations and Transport in Single-File Diffusive Systems

2025

- Computed current and tracer cumulants for one-dimensional diffusive particle systems using macroscopic fluctuation theory.
- Derived transport-coefficient transformations connecting Brownian particles, hard rods, and exclusion processes.

Spin Transport in the Heisenberg Spin Chain

2024

- Computed quantum response functions analytically and numerically using exact diagonalisation techniques.
- Investigated KPZ universality in spin transport using generalized hydrodynamics and thermodynamic Bethe ansatz methods.
- Built numerical simulation workflows for quantum many-body systems in Python.

Quantum Walks and Quantum Algorithms

2025

- Studied continuous-time and discrete-time quantum walks and their applications in quantum algorithms.
- Analyzed quantum query complexity for NAND-tree evaluation problems and simulated quantum speedup on welded-tree and hierarchical graphs using Qiskit and Python.

TEACHING EXPERIENCE

Teaching Assistant — Bangalore School of Statistical Physics (BSSP)

Jul 2025

- Led tutorial sessions for graduate-level statistical physics lectures attended by PhD students and postdoctoral researchers across India.
- Assisted with derivations in stochastic processes, non-equilibrium statistical mechanics, hydrodynamics, and transport theory.
- Helped communicate complex theoretical concepts clearly to academically diverse audiences.

TECHNICAL SKILLS

Programming & Tools: Python, NumPy, SciPy, Matplotlib, Qiskit, PyTorch, $\text{\LaTeX}$, Gnuplot

Numerical Methods: Molecular Dynamics & Event-Driven Simulation, Monte Carlo, Exact Diagonalisation, Numerical ODE/PDE Solving, Finite-Size Scaling

Research Areas: Statistical Physics, Non-Equilibrium Systems, Active Matter, Quantum Many-Body Systems, Quantum Algorithms, Stochastic Processes, Generalized & Fluctuating Hydrodynamics, Scientific Computing

CONFERENCES, SCHOOLS & PRESENTATIONS

- Indian Statistical Physics Community Meeting (ISPCM 2026), Bengaluru — Poster Presentation
- Bangalore School of Statistical Physics (BSSP) 2025 — Participant and Teaching Assistant
- Bangalore School of Statistical Physics (BSSP) 2024 — Participant

HONORS & ACHIEVEMENTS

- INSPIRE Scholar — Department of Science and Technology, Government of India
- JEST Physics 2023 — All India Rank **8**
- IIT JAM Physics 2023 — All India Rank **9**
- JEE Advanced 2020 — Qualified
- NTSE 2018 — Stage-I Qualified, State Rank 4 (Assam)

LANGUAGES

- **English** — Native / Full Professional Proficiency
- **Bangla** — Native / Full Professional Proficiency
- **Assamese** — Advanced
- **Hindi** — Advanced